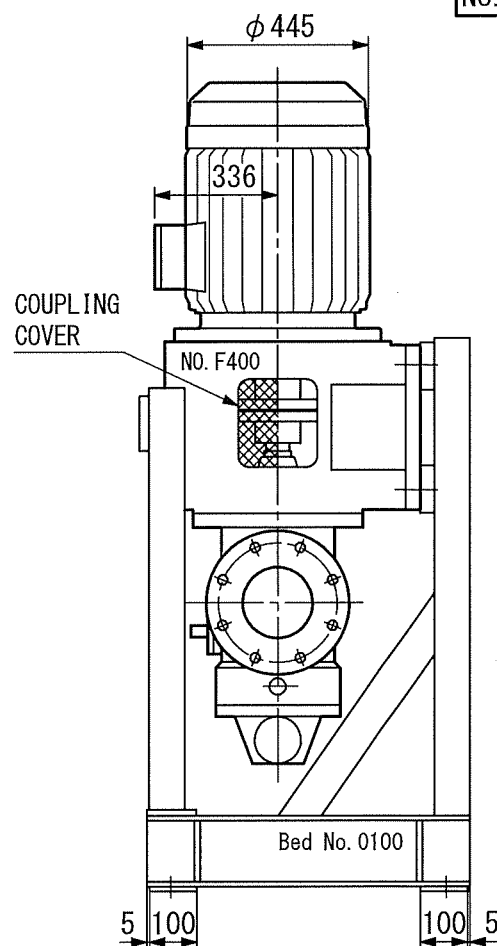
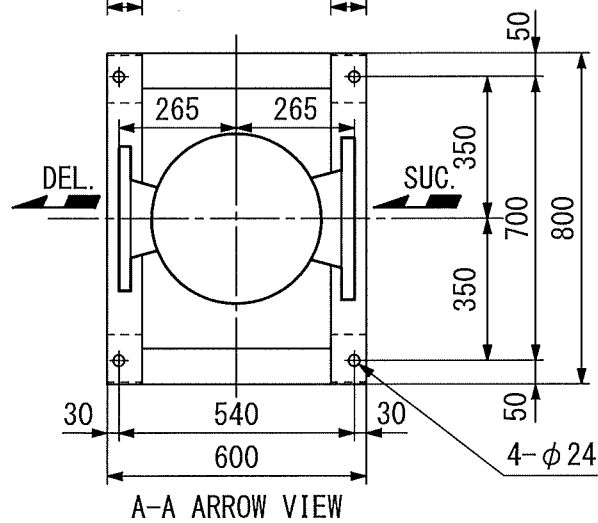
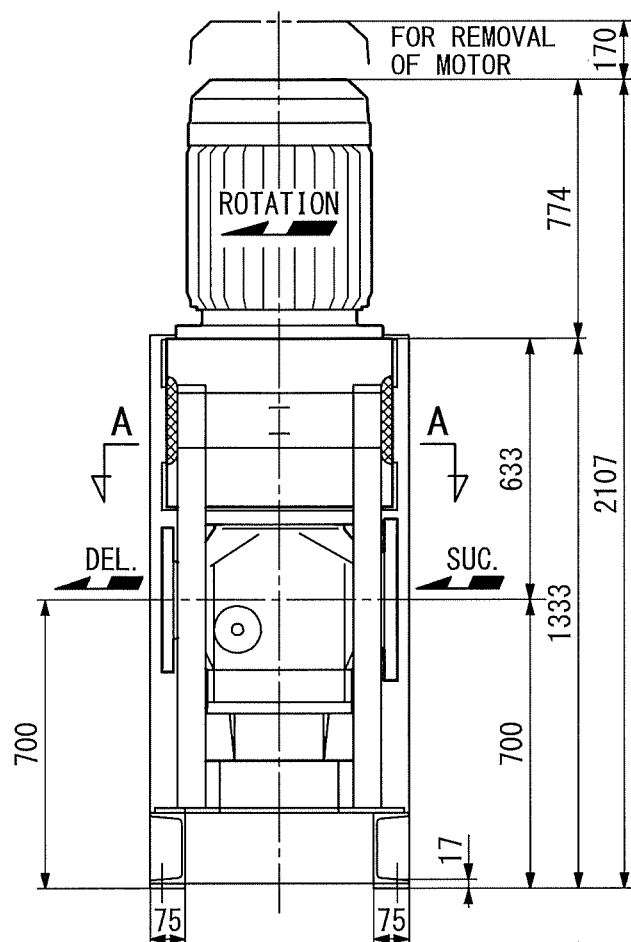




SUC. & DEL. FLANGE

DIN PN16	DIN PN16
SUC.	DEL.
φ 200	φ 150
φ 295	φ 240
φ 340	φ 285
12-φ 22	8-φ 22

MASS	
PUMP	700 kg
MOTOR	310 kg
TOTAL	1010 kg

ACCESSORIES FOR EACH 1 PUMP		SPECIFICATION		PUMP NAME	
DESCRIPTION	Q' TY	SUCTION BORE	200 mm	ULSHFO TRANSFER PUMP FUEL OIL CONSTANT TRANSFER PUMP MGO TRANSFER PUMP	
COUPLING & COVER	1set	DELIVERY BORE	150 mm		
SAFETY VALVE (SET PRESSURE 0.45 MPa)	1set	CAPACITY	80 m ³ /h		
PRESSURE GAUGE WITH JOINT	1set	TOTAL PRESSURE	0.4 MPa		
COMPOUND GAUGE WITH JOINT	1set	SUCTION PRESSURE	-0.05 MPa	PUMP MODEL	ALTV-125
GAUGE ROOT VALVE	2	REVOLVING SPEED	1150 min ⁻¹	QUANTITY	TOTAL 3
GAUGE BOARD	1	HYD. TEST PRESS.	0.9 MPa	RULE	LR
COUNTER FLANGE	1set	MOTOR OUTPUT	30kW , 6P	ANGLE PROJ.	3 rd
		POWER SOURCE	440V, 60Hz, 3ph	SCALE	FREE
		GUARANT. VIS. FOR POWER	1000 mm ² /s	DRAWING NO.	C4DG0242
		GUARANT. VIS. FOR CAPA.	2 mm ² /s	NANIWA PUMP MFG.CO.,LTD. DESIGN SECTION TECHNICAL DEPARTMENT	
		OIL TEMP =	0~60 °C		



SECTIONAL DRAWING

VERTICAL 3-SCREW PUMP

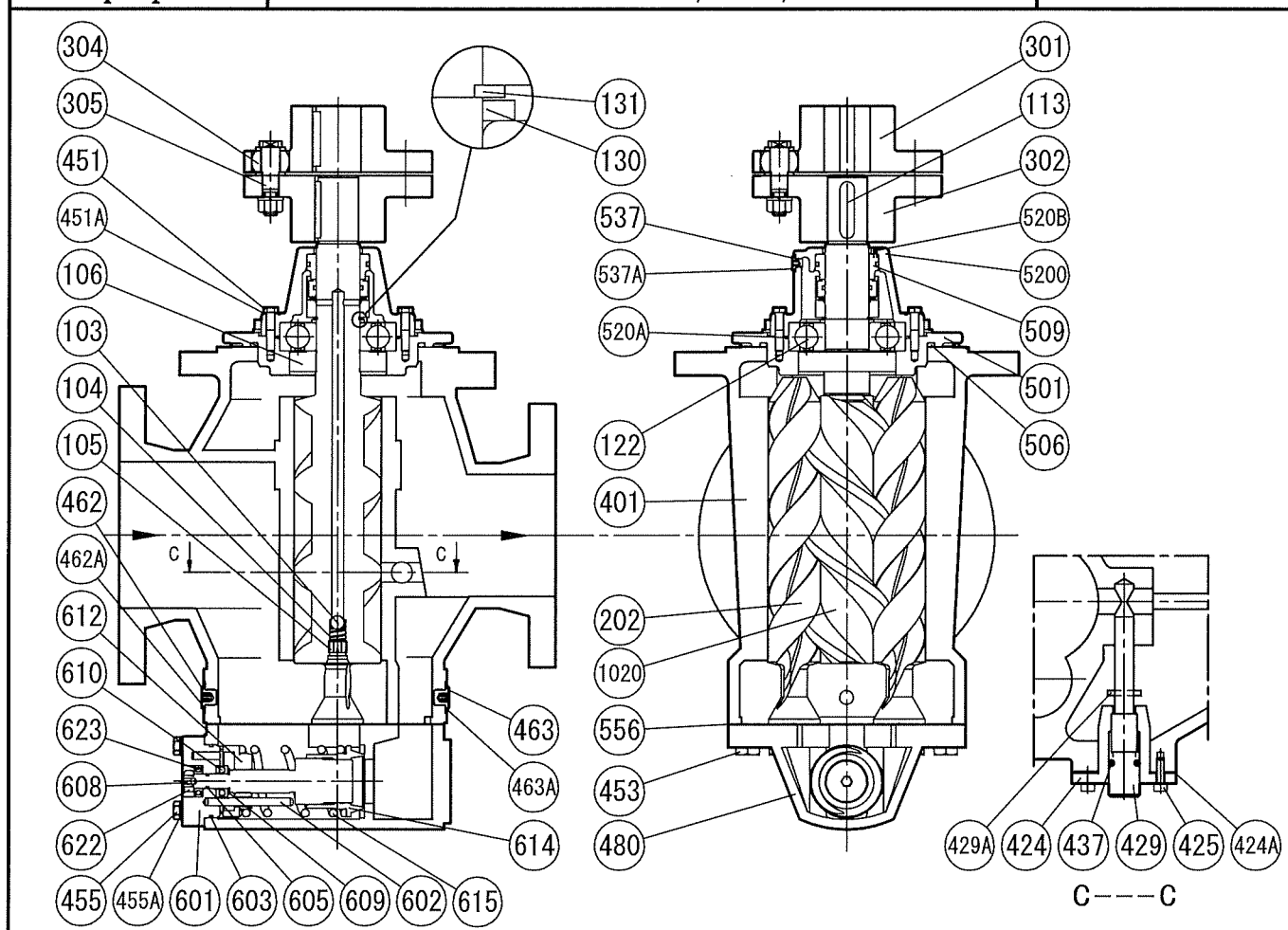
MODEL : ALTV-100/110/125

DRAWING NO.

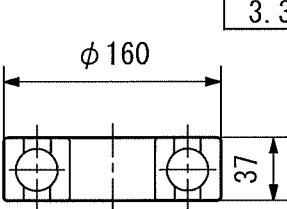
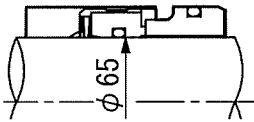
B9DS0058

DATE

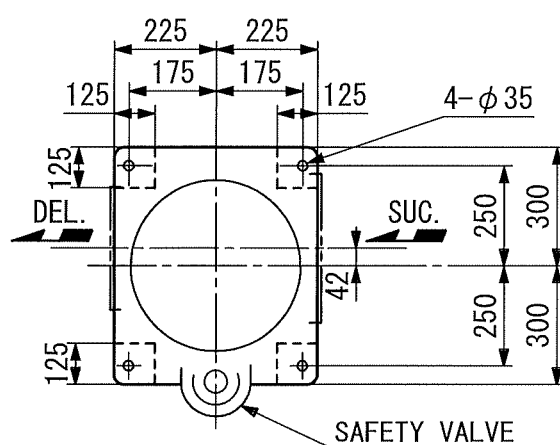
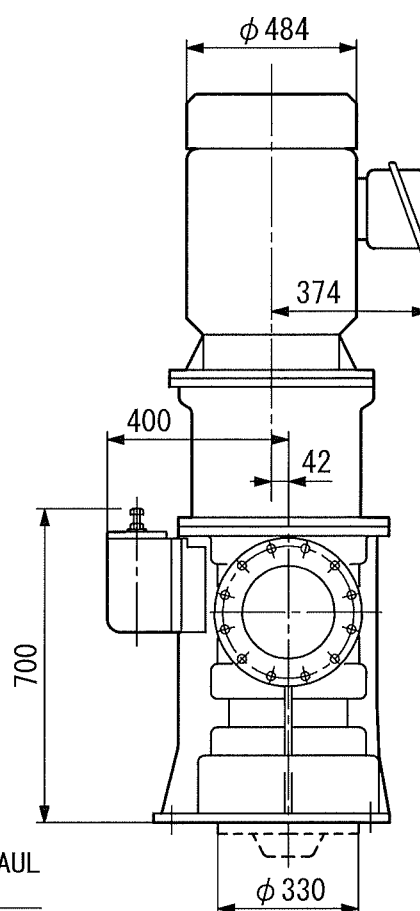
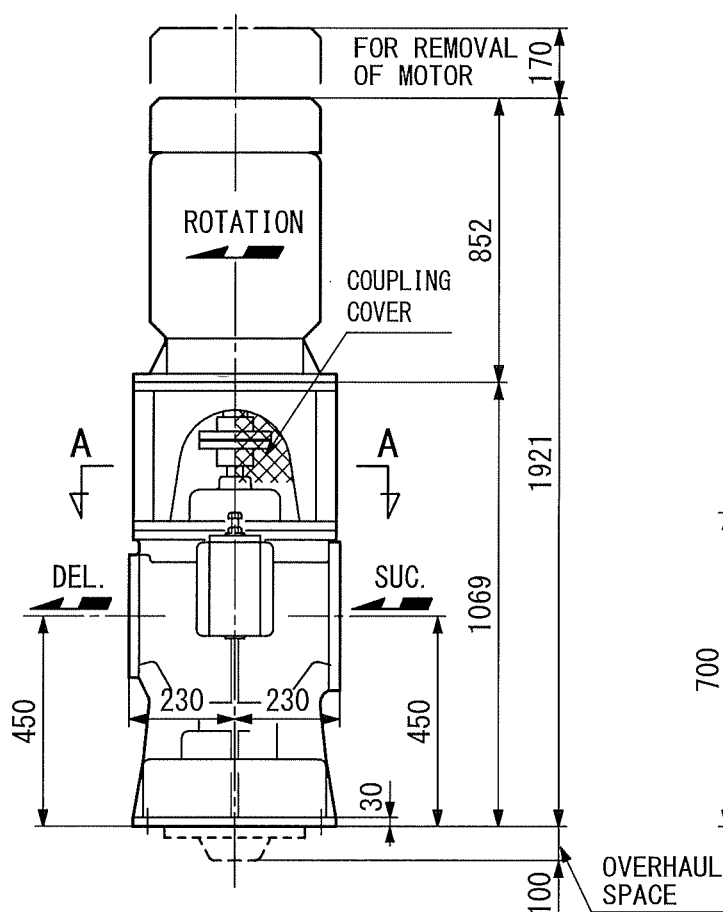
7 MAR. 2019



455A	WASHER	MILD STEEL	SS400	4	623	BALL BEARING	SUJ		1
455	SCREW	MILD STEEL	SS400	4	622	NUT	MILD STEEL	SS400	1
453	SCREW	MILD STEEL	SS400	8	615	VALVE SPRING	CARBON STEEL	S45C	1
451A	WASHER	MILD STEEL	SS400	4	614	VALVE PISTON	CARBON STEEL	S45C	1
451	SCREW	MILD STEEL	SS400	4	612	REGULATING NUT	MILD STEEL	SS400	1
437	O RING	RUBBER		1	610	BALL BEARING	SUJ		1
429A	TENSION PIN	CARBON STEEL	S45C	1	609	WASHER	MILD STEEL	SS400	1
429	GUIDING SCREW	CARBON STEEL	S45C	1	608	VALVE SPINDLE	MILD STEEL	SS400	1
425	SCREW	MILD STEEL	SS400	4	605	O RING	RUBBER		1
424A	GASKET	KLINGER SIL		1	603	O RING	RUBBER		1
424	COVER	CARBON STEEL	S45C	1	602	PIN	CARBON STEEL	S45C	1
401	PUMP BODY	CAST IRON	FC200	1	601	VALVE COVER	DUCTILE CAST IRON	FCD400	1
305	COUPLING BOLT	MILD STEEL	SS400	1SET	556	GASKET	KLINGER SIL		1
304	COUPLING RING	RUBBER		1SET	537A	WASHER	ALUMINIUM		1
302	COUPLING	CAST IRON	FC200	1	537	PLUG	MILD STEEL	SS400	1
301	COUPLING	CAST IRON	FC200	1	520B	TENSION PIN	STEEL		1
202	IDLER ROTOR	STRUCTURAL STEEL	EN10025-2 S450J0	2	520A	O RING	RUBBER		1
131	RETAINING RING	CARBON STEEL	S45C	1	5200	COMPL. COVER	DUCTILE CAST IRON	FCD400	1
130	SUPPORT RING	CARBON STEEL	S45C	1	509	MECHANICAL SEAL			1SET
122	BALL BEARING	SUJ		1	506	O RING	RUBBER		1
113	KEY	CARBON STEEL	S45C	1	501	FRONT COVER	DUCTILE CAST IRON	FCD400	1
106	BALANCING PISTON	CARBON STEEL	S45C	1	480	VALVE HOUSING	DUCTILE CAST IRON	FCD400	1
105	HOLE SCREW	MILD STEEL	SS400	1	463A	WASHER	ALUMINIUM		1
104	SPRING	CARBON STEEL	S45C	1	463	PLUG	MILD STEEL	SS400	1
103	BALL	CARBON STEEL	S45C	1	462A	WASHER	ALUMINIUM		1
1020	COMPL. POWER ROTOR	STRUCTURAL STEEL	EN10025-2 S450J0	1	462	PLUG	MILD STEEL	SS400	1
PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY

SPARE PARTS LIST							PAGE		1, 2, 3		
ULSHFO TRANSFER PUMP FUEL OIL CONSTANT TRANSFER PUMP MGO TRANSFER PUMP							VERTICAL 3-SCREW PUMP MODEL: <u>ALTV-125</u> TOTAL 3 SETS				
							SHIP NO.				
							BOX NO.				
Item NO.	NAME	SKETCH	MASS : kg	MATERIAL	SUPPLY		DRAWING NO.		REMARKS		
					WORK- ING PER PUMP	SPARE PER SHIP	NO.	PART NO.			
1	BALL BEARING		3.3	SUJ	1	1		122			
2	MECHANICAL SEAL		0.65		1 SET	1 SET		509			

MFR'S NAME & ADDRESS	 NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN	DRAWING NO.
		B6DP0029



A-A ARROW VIEW

SUC. & DEL. FLANGE

JIS 10K SUC.	JIS 16K DEL.
φ 200	φ 150
φ 290	φ 260
φ 330	φ 305
12-φ 23	12-φ 25

MASS

PUMP	380 Kg
MOTOR	415 Kg
TOTAL	795 Kg

ACCESSORIES FOR EACH 1 PUMP		SPECIFICATION		PUMP NAME	
DESCRIPTION	Q' TY	SUCTION BORE	200 mm	NO. 1 & NO. 2 M/E CROSSHEAD L. O. PUMP	
COUPLING & COVER	1set	DELIVERY BORE	150 mm		
AIR COCK	1	CAPACITY	100 m ³ /h	PUMP MODEL	ALSV-118N
SAFETY VALVE (SET PRESSURE 0.8 MPa)	1set	TOTAL PRESSURE	0.75 MPa	QUANTITY	2
PRESSURE GAUGE WITH JOINT	2sets	SUCTION PRESSURE	+0.69 MPa	RULE	LR
GAUGE ROOT VALVE	2	REVOLVING SPEED	1750 min ⁻¹	ANGLE PROJ.	3 rd
GAUGE BOARD	1	HYD. TEST PRESS.	2.5 MPa	SCALE	FREE
COUNTER FLANGE	1set	MOTOR OUTPUT	55kW, 4P	DRAWING NO.	B4DG0154
		POWER SOURCE	440V, 60Hz, 3ph	NANIWA PUMP MFG.CO.,LTD. DESIGN SECTION TECHNICAL DEPARTMENT	
		GUARANT. VIS. FOR POWER	260 mm ² /s		
		GUARANT. VIS. FOR CAPA.	75 mm ² /s		
		OIL TEMP = 60 °C			



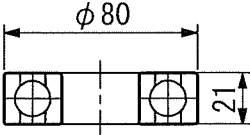
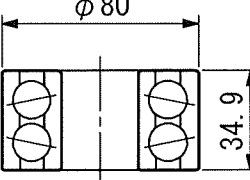
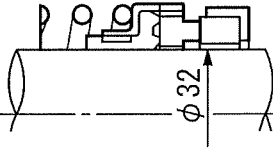
PARTS LIST

VERTICAL 2-SCREW PUMP

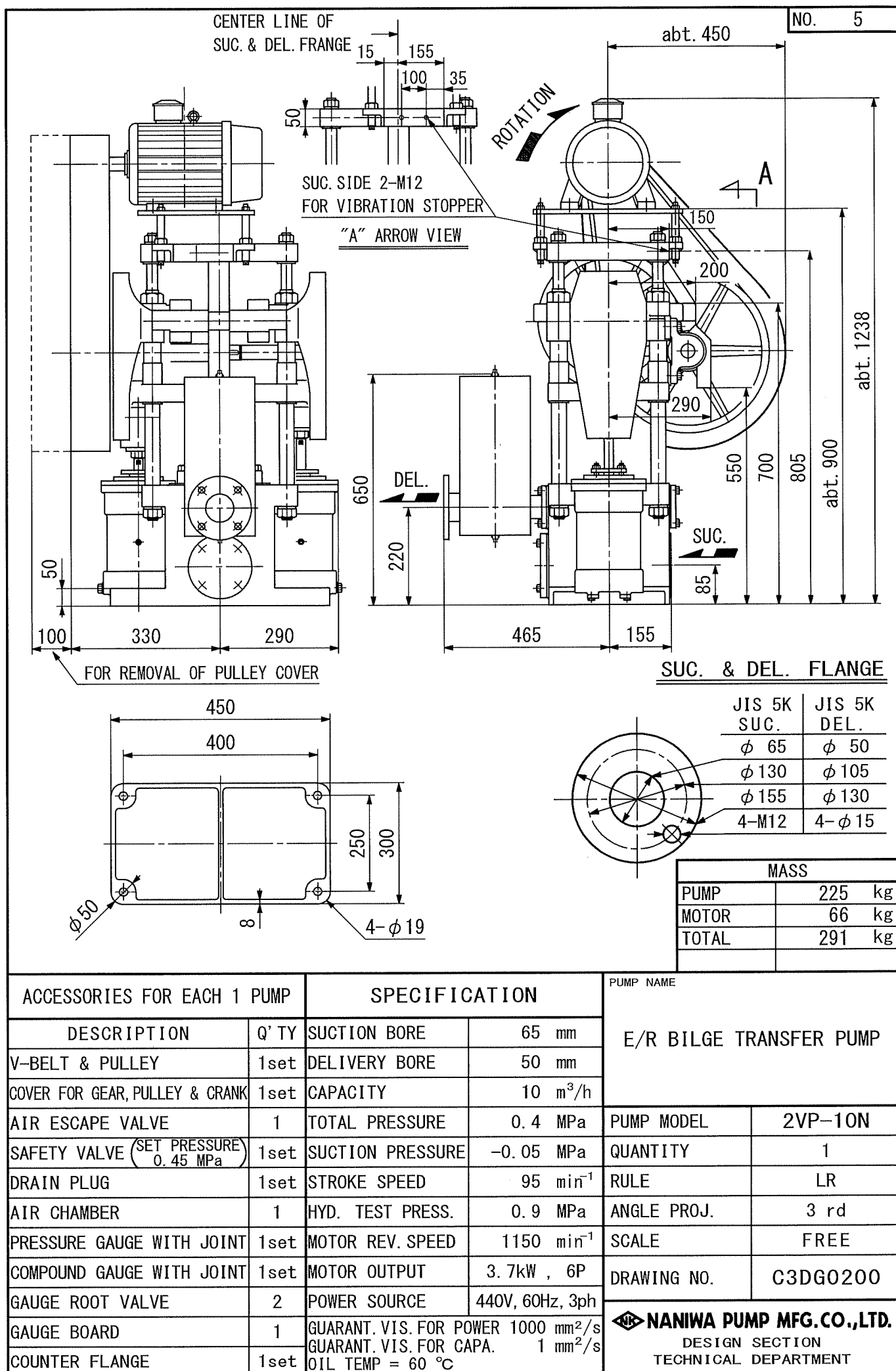
MODEL ALSV-118N

DRAWING NO.
B5DS0242
DATE
31 JUL. 2015

PART NO.	NAME OF PART	MATERIAL	Q' TY	PART NO.	NAME OF PART	MATERIAL	Q' TY
101	SHORT SCREW SHAFT	CARBON STEEL S45C	1	494	SEGMENT	CAST IRON FC200	2
102	LONG SCREW SHAFT	CARBON STEEL S45C	1	702	MECHANICAL SEAL		1 SET
103	BALL BEARING	SUJ #6307	2	803	MOTOR FRAME	CAST IRON FC200	1
105	BALL BEARING	SUJ #3307	2	804	HEXAGON HEAD BOLT	MILD STEEL SS400	8
108	TIMING GEAR	CHROMIUM MOLYB. STEEL SCM420H	2	900	VALVE CASING	DUCTILE CAST IRON FCD400	1
110	BEARING NUT	MILD STEEL SS400	2	901	VALVE COVER	DUCTILE CAST IRON FCD400	1
111	BEARING WASHER	MILD STEEL SS400	2	904	VALVE CONE	CAST IRON FC200	1
124	DISTANCE BUSH	CARBON STEEL S45C	2	905	SPRING SEAT	MILD STEEL SS400	1
130	SHIM	COPPER ALLOY ROD C2801P	2	907	ADJUSTING BOLT	MILD STEEL SS400	1
131	SHIM	COPPER ALLOY ROD C2801P	2	908	HEXAGON HEAD NUT	MILD STEEL SS400	1
132	TIMING GEAR KEY	CARBON STEEL S45C	2	910	VALVE SPRING	SPRING STEEL SUP6	1
133	COUPLING KEY	CARBON STEEL S45C	1	918	HEXAGON HEAD BOLT	MILD STEEL SS400	4
301	COUPLING	CAST IRON FC200	1	920	JOINT PACKING	NON ASBESTOS FIBER	1
302	COUPLING	CAST IRON FC200	1	925	O RING	RUBBER	1
304	COUPLING RING	RUBBER	1 SET	926	O RING	RUBBER	1
305	COUPLING BOLT & NUT	MILD STEEL SS400	1 SET	927	O RING	RUBBER	1
310	HEXAGON HEAD BOLT	MILD STEEL SS400	8	930	PLUG	COPPER ALLOY ROD C3602BD	1
350	PUMP CASING	DUCTILE CAST IRON FCD400	1	931	JOINT PACKING	FIBER	1
375	FITTED PIN	CARBON STEEL S45C	4				
376	FITTED PIN	CARBON STEEL S45C	2				
401	COVER "DF"	DUCTILE CAST IRON FCD400	1				
406	GEAR BOX	CAST IRON FCD400	1				
407	COVER "D"	DUCTILE CAST IRON FCD400	1				
422	PLUG	COPPER ALLOY ROD C3602BD	1				
434	HEXAGON HEAD BOLT	MILD STEEL SS400	8				
439	JOINT PACKING	NON ASBESTOS FIBER	2				
440	JOINT PACKING	NON ASBESTOS FIBER	1				
443	PLUG	COPPER ALLOY ROD C3602BD	1				
471	HEXAGON HEAD BOLT	CARBON STEEL S45C	8				
472	TONGUED WASHER	MILD STEEL SS400	8				

SPARE PARTS LIST							PAGE		4		
NO. 1 & NO. 2 M/E CROSSHEAD L. O. PUMP							VERTICAL 2-SCREW PUMP MODEL: ALSV-118N 2 SETS				
							SHIP NO.				
							BOX NO.				
Item NO.	NAME	SKETCH	MASS : kg	MATERIAL	SUPPLY		DRAWING NO.		REMARKS		
					WORK- ING PER PUMP	SPARE PER SHIP	B5DS0242				
							NO.	PART NO.			
1	BALL BEARING		0.46	SAE52100	2	1		103			
2	BALL BEARING		0.79	SAE52100	2	1		105			
3	MECHANICAL SEAL		0.09		1 SET	1 SET		702			

MFR' S NAME & ADDRESS	 NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN	DRAWING NO.
		99DP0391





MATERIAL LIST

VERTICAL PISTON PUMP

MODEL : 2VP-N

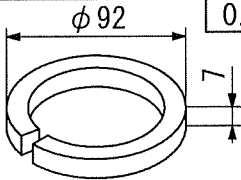
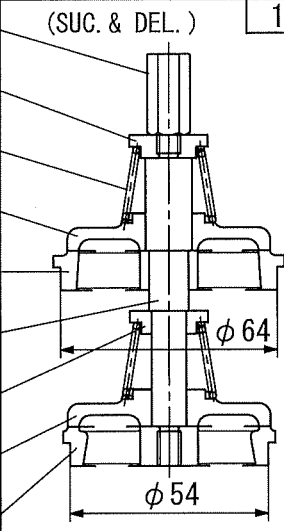
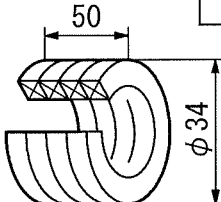
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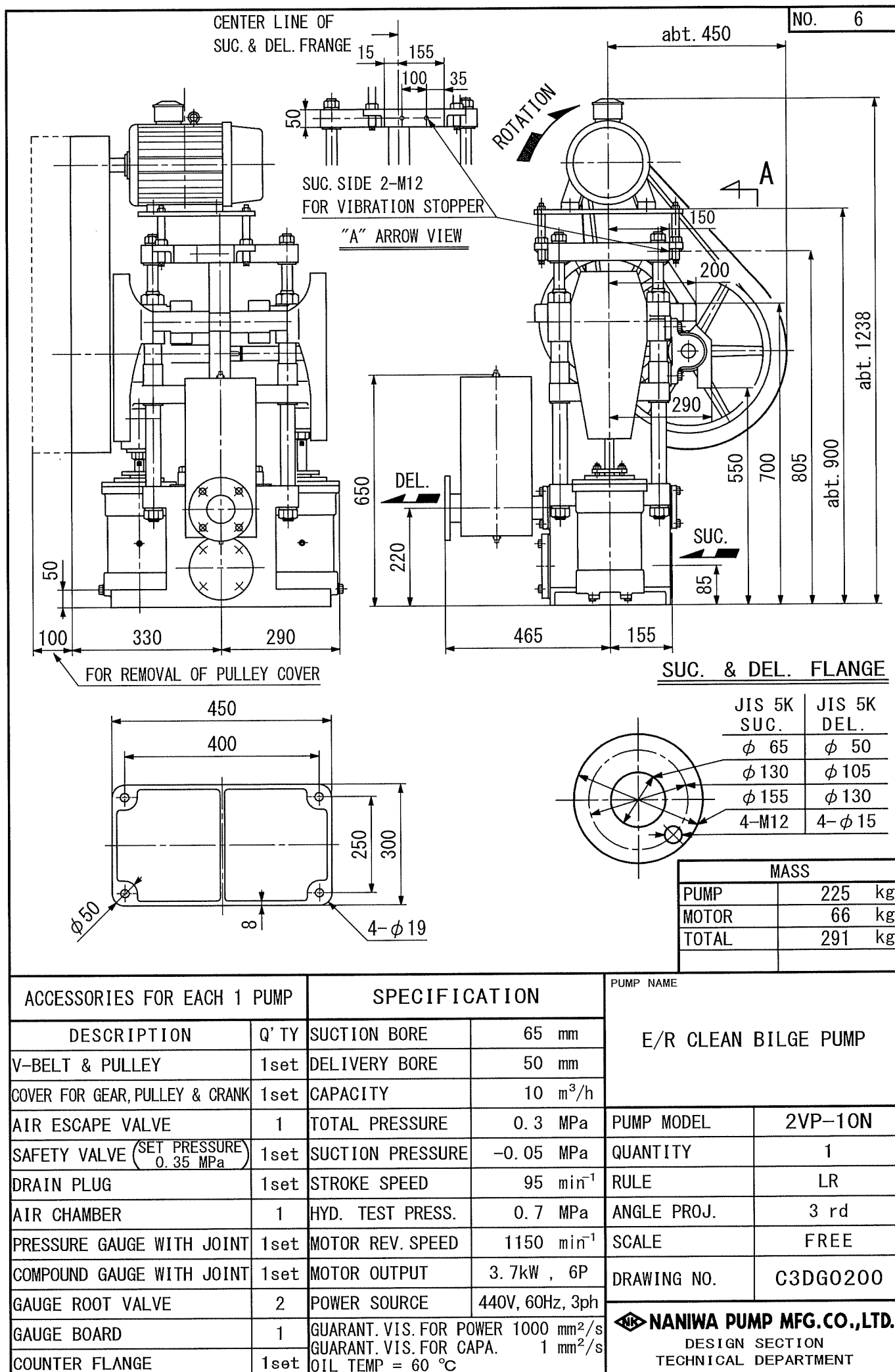
B3DS0294

DATE

9 AUG. 2013

PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY
101	PUMP CYLINDER	CAST IRON	FC200	1	319	WASHER	CARBON STEEL	SPCC	2
102	LINER	CAST BRONZE	CAC406	2	321	HEAD PIN METAL	CAST BRONZE	CAC406	2
103	CYLINDER COVER	CAST IRON	FC200	2	322	CONNECTING ROD	DUCTILE CAST IRON	FCD400	2
106	BOTTOM COVER	CAST IRON	FC200	2	331	GUIDE FRAME	CAST IRON	FC200	1
110	BUCKET	CAST BRONZE	CAC406	4	332	DISTANCE PIECE	CARBON STEEL TUBE	SGP	4
111	BUCKET RING	BAKELITE		4	333	SPECIAL NUT	MILD STEEL	SS400	4
112	DISTANCE RING	CAST BRONZE	CAC406	2	361	CRANK FRAME	CAST IRON	FC200	1
116	COVER PUSH	CAST IRON	FC200	1	362	MOTOR FRAME	CAST IRON	FC200	1
160 1	CIRCLIP	CARBON TOOL STEEL	SK5	1	366	MOTOR BASE	MILD STEEL	SS400	1
160 2	CIRCLIP	CARBON TOOL STEEL	SK5	2	501	GLAND PACKING	CARBONIZED FIBER		2 SETS
160 3	CIRCLIP	CARBON TOOL STEEL	SK5	4	504	GLAND	CAST BRONZE	CAC406	2
160 4	CIRCLIP	CARBON TOOL STEEL	SK5	1	601	CRANK ARM SHAFT	CARBON STEEL	S45C	1
160 5	CIRCLIP	CARBON TOOL STEEL	SK5	4	602	COUNTER SHAFT	CHROM. MOLYB. STEEL	SCM435	1
161 1	CIRCLIP	CARBON TOOL STEEL	SK5	2	603	CRANK PIN	CARBON STEEL	S45C	2
174 1	SPLIT PIN	LOW CARBON STEEL WIRE	SWRM	2	604	CRANK ARM	DUCTILE CAST IRON	FCD400	2
174 2	SPLIT PIN	LOW CARBON STEEL WIRE	SWRM	2	605	ARM COLLAR	MILD STEEL	SS400	2
210 1	SUCTION VALVE	CAST BRONZE	CAC406	4	606	GEAR COLLAR	MILD STEEL	SS400	1
210 2	DELIVERY VALVE	CAST BRONZE	CAC406	4	612	ARM KEY	CARBON STEEL	S45C	2
212	LIFT BOLT NUT	STAINLESS STEEL	SUS304	4	613	GEAR KEY	CARBON STEEL	S45C	1
213	DELIVERY VALVE SEAT	CAST BRONZE	CAC406	4	630	BALL BEARING	SUJ		2
214	SUCTION VALVE SEAT	CAST BRONZE	CAC406	4	631	BALL BEARING	SUJ		4
215	SPRING	PHOSPHOR BRONZE WIRE	PBW	8	633	COUNTER METAL	SUJ		2
216	LIFT BOLT	STAINLESS STEEL	SUS304	4	662	PINION GEAR	CARBON STEEL	S45C	1
217	SPRING SEAT (DELIVERY)	STAINLESS STEEL	SUS304	4	664	GEAR	CAST IRON	FC200	1
218	SPRING SEAT (SUCTION)	STAINLESS STEEL	SUS304	4	674	V PULLEY (MOTOR SIDE)	CAST IRON	FC200	1
221	SET BOLT	MILD STEEL	SS400	4	675	V PULLEY (PUMP SIDE)	CAST IRON	FC200	1
225	FLANGE COVER	CAST IRON	FC200	1	676	V BELT			4
226	FLANGE COVER	MILD STEEL	SS400	1	677	PINION GEAR KEY	CARBON STEEL	S45C	1
230	VALVE COVER	CAST IRON	FC200	4	678	PULLEY KEY	CARBON STEEL	S45C	1
232	O RING	RUBBER		4	702	COVER	CAST IRON	FC200	1
233	O RING	RUBBER		1	704	VALVE	CAST BRONZE	CAC406	1
302	PUMP ROD	STAINLESS STEEL	SUS304	2	707	SPRING	STAINLESS STEEL	SUS-WP	1
305	CROSS HEAD	CAST IRON	FC200	2	710	SPRING SEAT	COPPER ALLOY ROD	C3604	1
308	STAY	MILD STEEL	SS400	4	713	ADJUST BOLT	STAINLESS STEEL	SUS304	1
311	CROSS HEAD PIN	CARBON STEEL	S45C	2	850	AIR CHAMBER	CARBON STEEL TUBE	SGP	1
315	PUMP ROD NUT	COPPER ALLOY ROD	C3604	2	855	GREASE CUP			2
316	NUT	COPPER ALLOY ROD	C3604	2					

SPARE PARTS LIST							PAGE		5, 6		
E/R BILGE TRANSFER PUMP E/R CLEAN BILGE PUMP							VERTICAL PISTON PUMP MODEL : 2VP-10N				
							TOTAL 2 SETS				
Item NO.	NAME	SKETCH	MASS : kg	MATERIAL	SUPPLY		DRAWING NO. B3DS0294		REMARKS		
					WORK- ING PER PUMP	SPARE PER SHIP	NO.	PART NO.			
1	BUCKET RING		0.01	BAKELITE	4	2		111			
2	LIFT BOLT NUT		1.0	STAINLESS STEEL	4 SETS	1 SET		212			
	DEL. SPRING SEAT			STAINLESS STEEL				217			
	VALVE SPRING			PHOSPHOR BRONZE				215			
	DEL. VALVE			CAST BRONZE				210/2			
	DEL. VALVE SEAT			CAST BRONZE				213			
	LIFT BOLT			STAINLESS STEEL				216			
	SUC. SPRING SEAT			STAINLESS STEEL				218			
	SUC. VALVE			CAST BRONZE				210/1			
	SUC. VALVE SEAT			CAST BRONZE				214			
3	GLAND PACKING		0.1	CARBONIZED FIBER	2 SETS	1 SET		501			
MFR' S NAME & ADDRESS		 NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN						DRAWING NO. C3DP0059			





MATERIAL LIST

VERTICAL PISTON PUMP

MODEL : 2VP-N

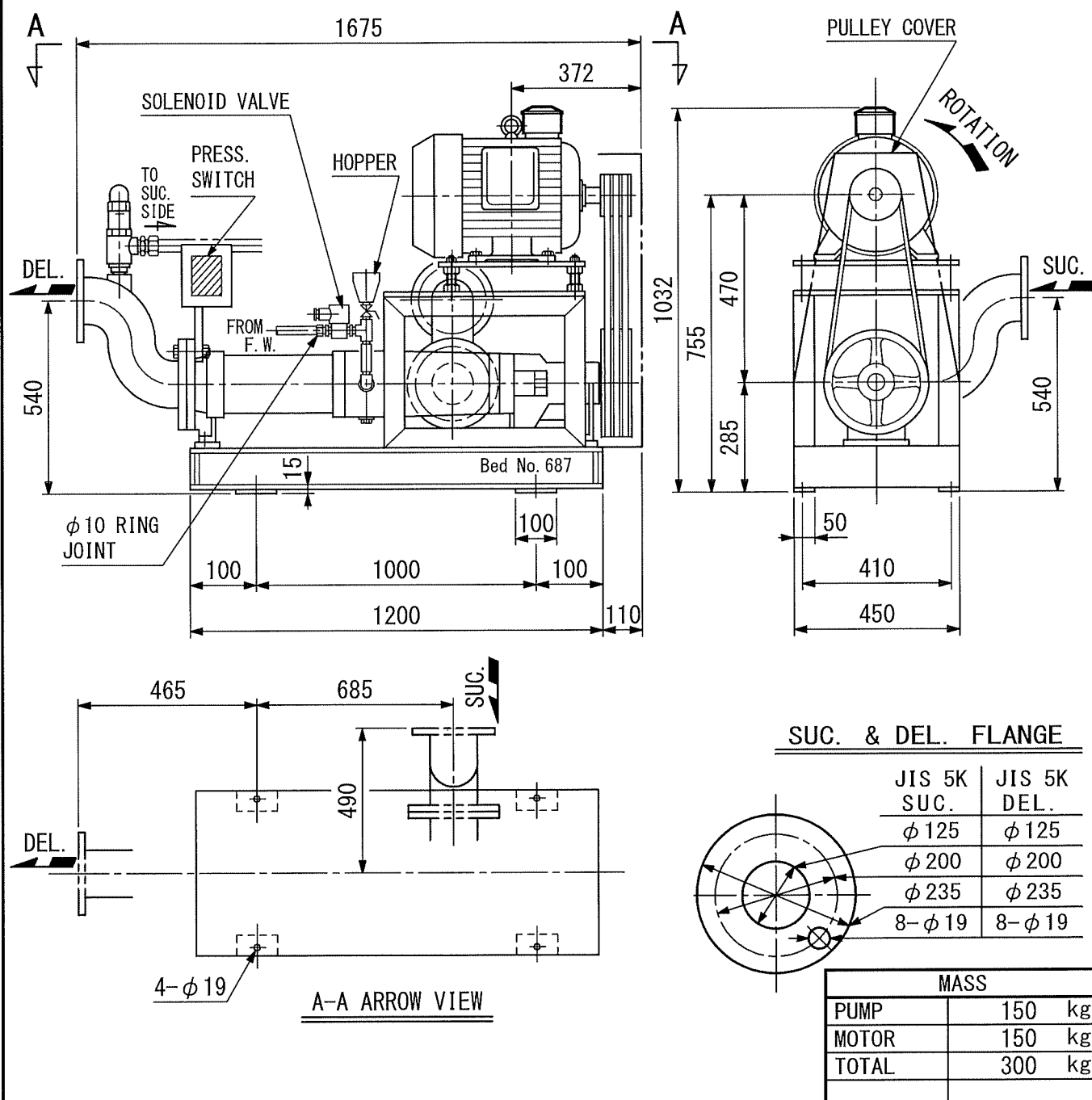
DRAWING NO.

B3DS0294

DATE

9 AUG. 2013

PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY
101	PUMP CYLINDER	CAST IRON	FC200	1	319	WASHER	CARBON STEEL	SPCC	2
102	LINER	CAST BRONZE	CAC406	2	321	HEAD PIN METAL	CAST BRONZE	CAC406	2
103	CYLINDER COVER	CAST IRON	FC200	2	322	CONNECTING ROD	DUCTILE CAST IRON	FCD400	2
106	BOTTOM COVER	CAST IRON	FC200	2	331	GUIDE FRAME	CAST IRON	FC200	1
110	BUCKET	CAST BRONZE	CAC406	4	332	DISTANCE PIECE	CARBON STEEL TUBE	SGP	4
111	BUCKET RING	BAKELITE		4	333	SPECIAL NUT	MILD STEEL	SS400	4
112	DISTANCE RING	CAST BRONZE	CAC406	2	361	CRANK FRAME	CAST IRON	FC200	1
116	COVER PUSH	CAST IRON	FC200	1	362	MOTOR FRAME	CAST IRON	FC200	1
160 1	CIRCLIP	CARBON TOOL STEEL	SK5	1	366	MOTOR BASE	MILD STEEL	SS400	1
160 2	CIRCLIP	CARBON TOOL STEEL	SK5	2	501	GLAND PACKING	CARBONIZED FIBER		2 SETS
160 3	CIRCLIP	CARBON TOOL STEEL	SK5	4	504	GLAND	CAST BRONZE	CAC406	2
160 4	CIRCLIP	CARBON TOOL STEEL	SK5	1	601	CRANK ARM SHAFT	CARBON STEEL	S45C	1
160 5	CIRCLIP	CARBON TOOL STEEL	SK5	4	602	COUNTER SHAFT	CHROM. MOLYB. STEEL	SCM435	1
161 1	CIRCLIP	CARBON TOOL STEEL	SK5	2	603	CRANK PIN	CARBON STEEL	S45C	2
174 1	SPLIT PIN	LOW CARBON STEEL WIRE	SWRM	2	604	CRANK ARM	DUCTILE CAST IRON	FCD400	2
174 2	SPLIT PIN	LOW CARBON STEEL WIRE	SWRM	2	605	ARM COLLAR	MILD STEEL	SS400	2
210 1	SUCTION VALVE	CAST BRONZE	CAC406	4	606	GEAR COLLAR	MILD STEEL	SS400	1
210 2	DELIVERY VALVE	CAST BRONZE	CAC406	4	612	ARM KEY	CARBON STEEL	S45C	2
212	LIFT BOLT NUT	STAINLESS STEEL	SUS304	4	613	GEAR KEY	CARBON STEEL	S45C	1
213	DELIVERY VALVE SEAT	CAST BRONZE	CAC406	4	630	BALL BEARING	SUJ		2
214	SUCTION VALVE SEAT	CAST BRONZE	CAC406	4	631	BALL BEARING	SUJ		4
215	SPRING	PHOSPHOR BRONZE WIRE	PBW	8	633	COUNTER METAL	SUJ		2
216	LIFT BOLT	STAINLESS STEEL	SUS304	4	662	PINION GEAR	CARBON STEEL	S45C	1
217	SPRING SEAT (DELIVERY)	STAINLESS STEEL	SUS304	4	664	GEAR	CAST IRON	FC200	1
218	SPRING SEAT (SUCTION)	STAINLESS STEEL	SUS304	4	674	V PULLEY (MOTOR SIDE)	CAST IRON	FC200	1
221	SET BOLT	MILD STEEL	SS400	4	675	V PULLEY (PUMP SIDE)	CAST IRON	FC200	1
225	FLANGE COVER	CAST IRON	FC200	1	676	V BELT			4
226	FLANGE COVER	MILD STEEL	SS400	1	677	PINION GEAR KEY	CARBON STEEL	S45C	1
230	VALVE COVER	CAST IRON	FC200	4	678	PULLEY KEY	CARBON STEEL	S45C	1
232	O RING	RUBBER		4	702	COVER	CAST IRON	FC200	1
233	O RING	RUBBER		1	704	VALVE	CAST BRONZE	CAC406	1
302	PUMP ROD	STAINLESS STEEL	SUS304	2	707	SPRING	STAINLESS STEEL	SUS-WP	1
305	CROSS HEAD	CAST IRON	FC200	2	710	SPRING SEAT	COPPER ALLOY ROD	C3604	1
308	STAY	MILD STEEL	SS400	4	713	ADJUST BOLT	STAINLESS STEEL	SUS304	1
311	CROSS HEAD PIN	CARBON STEEL	S45C	2	850	AIR CHAMBER	CARBON STEEL TUBE	SGP	1
315	PUMP ROD NUT	COPPER ALLOY ROD	C3604	2	855	GREASE CUP			2
316	NUT	COPPER ALLOY ROD	C3604	2					



ACCESSORIES FOR EACH 1 PUMP		SPECIFICATION		PUMP NAME	
DESCRIPTION	Q' TY	SUCTION BORE	125 mm	SLUDGE PUMP	
V-BELT & PULLEY & COVER	1set	DELIVERY BORE	125 mm		
COMMON BED	1	CAPACITY	30 m ³ /h		
SAFETY VALVE (SET PRESSURE 0.5 MPa)	1set	TOTAL PRESSURE	0.45 MPa	PUMP MODEL	AE1E 750-ID
HOPPER & VALVE	1set	SUCTION PRESSURE	-0.05 MPa	QUANTITY	1
PRESSURE GAUGE WITH JOINT	1set	REVOLVING SPEED	380 min ⁻¹	RULE	LR
COMPOUND GAUGE WITH JOINT	1set	HYD. TEST PRESS.	1.0 MPa	ANGLE PROJ.	3 rd
GAUGE BOARD & ROOT VALVE	1set	MOTOR REV. SPEED	1150 min ⁻¹	SCALE	FREE
SOLENOID VALVE	1set	MOTOR OUTPUT	11kW, 6P	DRAWING NO.	B2DGO479
PIECE (SUC. /DEL.)	1set	POWER SOURCE	440V, 60Hz, 3ph	NANIWA PUMP MFG.CO.,LTD. DESIGN SECTION TECHNICAL DEPARTMENT	
PRESS. SWITCH ON : 0.4 MPa OFF : 0.3 MPa	1	GUARANT. VIS. FOR POWER	1000 mm ² /s		
COUNTER FLANGE	1set	GUARANT. VIS. FOR CAPA.	26 mm ² /s		
		OIL TEMP = 80 °C			



SECTIONAL DRAWING

HORIZONTAL 1-SCREW PUMP

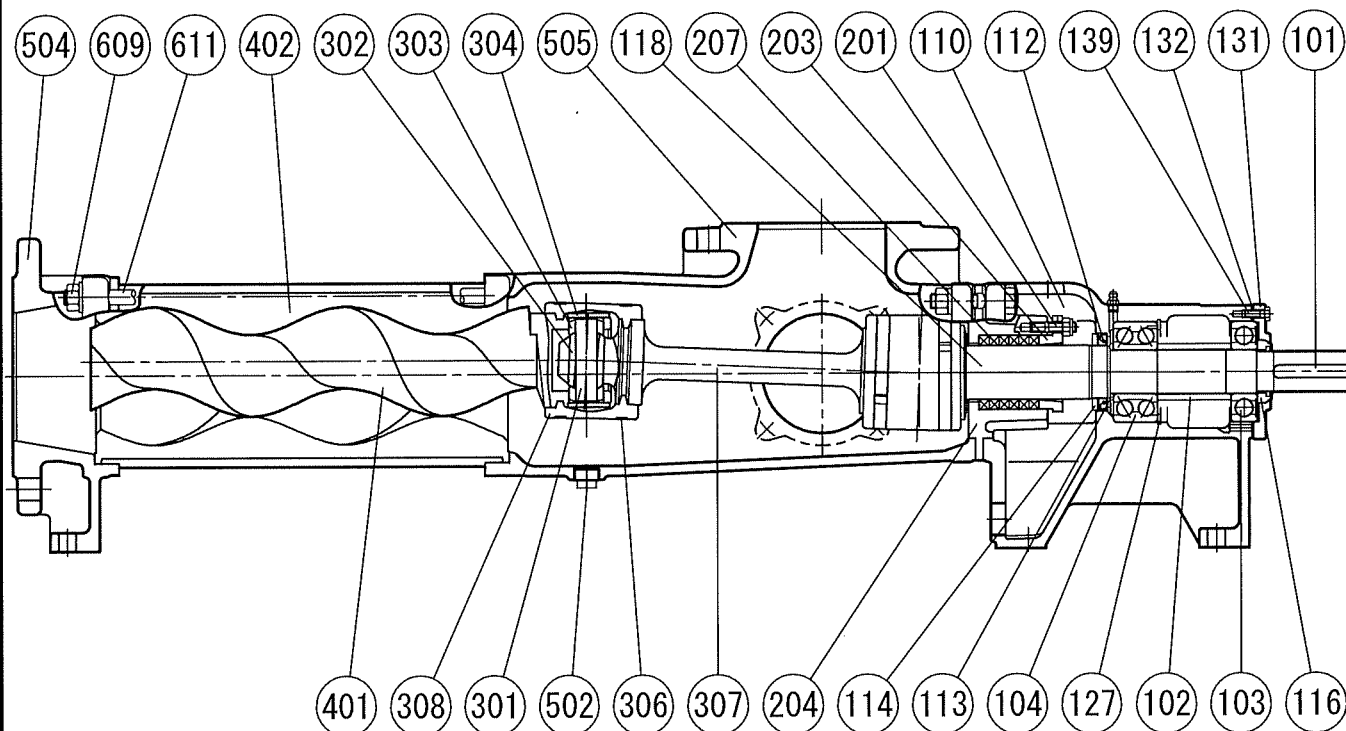
MODEL AE1E

DRAWING NO.

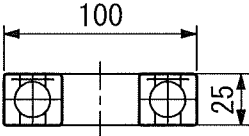
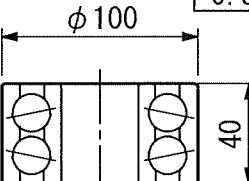
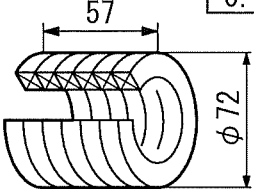
99DS0507

DATE

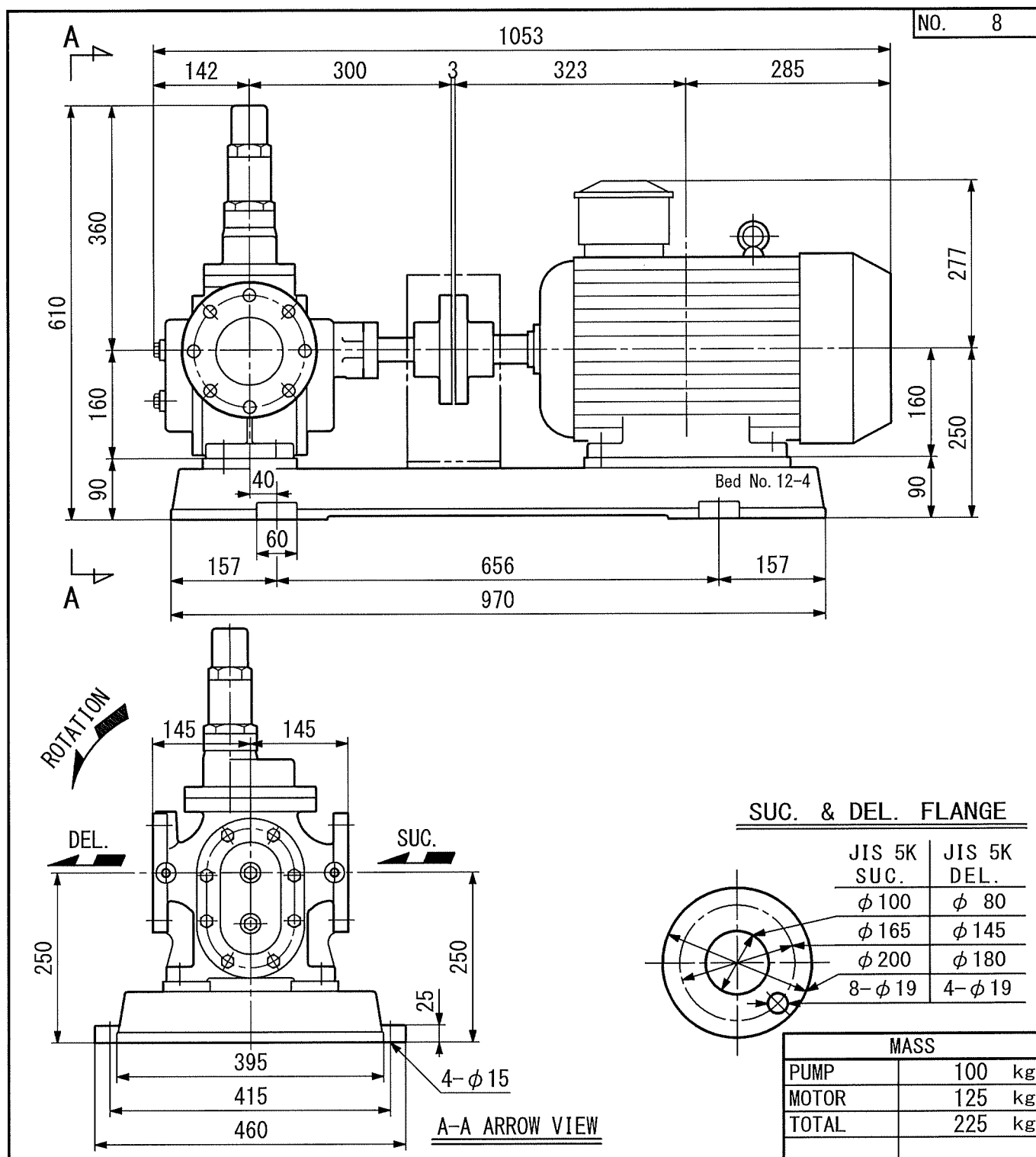
25 AUG. 1999



203	GLAND	CAST IRON	FC200	1	611	TIE BOLT	MILD STEEL	SS400	4
201	STUD BOLT	MILD STEEL	SS400	2	609	NUT	MILD STEEL	SS400	4
139	HEXAGON SCREW	MILD STEEL	SS400	3	505	SUCTION CASING	CAST IRON	FC200	1
132	GASKET			1	504	DISCHARGE CASING	CAST IRON	FC200	1
131	BEARING COVER	CAST IRON	FC200	1	502	PLUG	CAST BRONZE	CAC406	1
127	CIRCLIP	CARBON TOOL STEEL	SK5	1	402	STATOR	RUBBER		1
118	SHAFT	STAINLESS STEEL	SUS420	1	401	ROTOR	STAINLESS STEEL	SUS304	1
116	BEARING NUT	MILD STEEL	SS400	1	308	COVER SLEEVE	RUBBER		2
114	FLINGER	RUBBER		1	307	COUPLING ROD	STAINLESS STEEL	SUS420	1
113	SPACER RING	MILD STEEL	SS400	1	306	CLAMPING BAND	STAINLESS STEEL	SUS420	4
112	OIL SEAL	RUBBER		1	304	RETAINING	CARBON STEEL	S45C	2
110	BEARING HOUSING	CAST IRON	FC200	1	303	YOKE BUSH	CARBON STEEL	S45C	4
104	BALL BEARING	SUJ		1	302	COUPLING ROD BUSH	CARBON STEEL	S45C	2
103	BALL BEARING	SUJ		1	301	COUPLING ROD PIN	CARBON STEEL	S45C	2
102	SPACER SLEEVE	MILD STEEL	SS400	1	207	GLAND PACKING	CARBONIZED FIBER		1 SET
101	KEY	CARBON STEEL	S45C	1	204	GLAND HOUSING	CAST IRON	FC200	1
PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY

SPARE PARTS LIST							PAGE		7	
SLUDGE PUMP HORIZONTAL 1-SCREW PUMP MODEL: AE1E 750-ID 1 SET							SHIP NO.			
							BOX NO.			
Item NO.	NAME	SKETCH	MASS : kg	MATERIAL	SUPPLY		DRAWING NO.		REMARKS	
					WORK- ING PER PUMP	SPARE PER SHIP	99DS0507			
								NO.	PART NO.	
1	BALL BEARING		0.83	SUJ	1	1			103	
2	BALL BEARING		0.88	SUJ	1	1			104	
3	GLAND PACKING		0.16	CARBONIZED FIBER	1 SET	1 SET			207	

MFR' S NAME & ADDRESS	 NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN	DRAWING NO.
		A3DP0348



ACCESSORIES FOR EACH 1 PUMP		SPECIFICATION		PUMP NAME	
DESCRIPTION	Q' TY	SUCTION BORE	100 mm	LUBE OIL TRANSFER PUMP	
COUPLING (JIS φ 160)	1set	DELIVERY BORE	80 mm		
COUPLING COVER	1	CAPACITY	20 m ³ /h	PUMP MODEL	ALG-65
COMMON BED	1	TOTAL PRESSURE	0.45 MPa	QUANTITY	1
AIR COCK	1	SUCTION PRESSURE	-0.05 MPa	RULE	LR
SAFETY VALVE (SET PRESSURE 0.50 MPa)	1set	REVOLVING SPEED	1150 min ⁻¹	ANGLE PROJ.	3 rd
PRESSURE GAUGE WITH JOINT	1set	HYD. TEST PRESS.	1.0 MPa	SCALE	FREE
COMPOUND GAUGE WITH JOINT	1set	MOTOR OUTPUT	7.5kW , 6P	DRAWING NO.	C2DG0264
GAUGE ROOT VALVE	2	POWER SOURCE	440V, 60Hz, 3ph	NANIWA PUMP MFG.CO.,LTD. DESIGN SECTION TECHNICAL DEPARTMENT	
GAUGE BOARD	1	GUARANT. VIS. FOR POWER	1000 mm ² /s		
COUNTER FLANGE	1set	GUARANT. VIS. FOR CAPA.	26 mm ² /s		
		OIL TEMP	= 60 °C		



SECTIONAL DRAWING

HORIZONTAL GEAR ROTARY PUMP

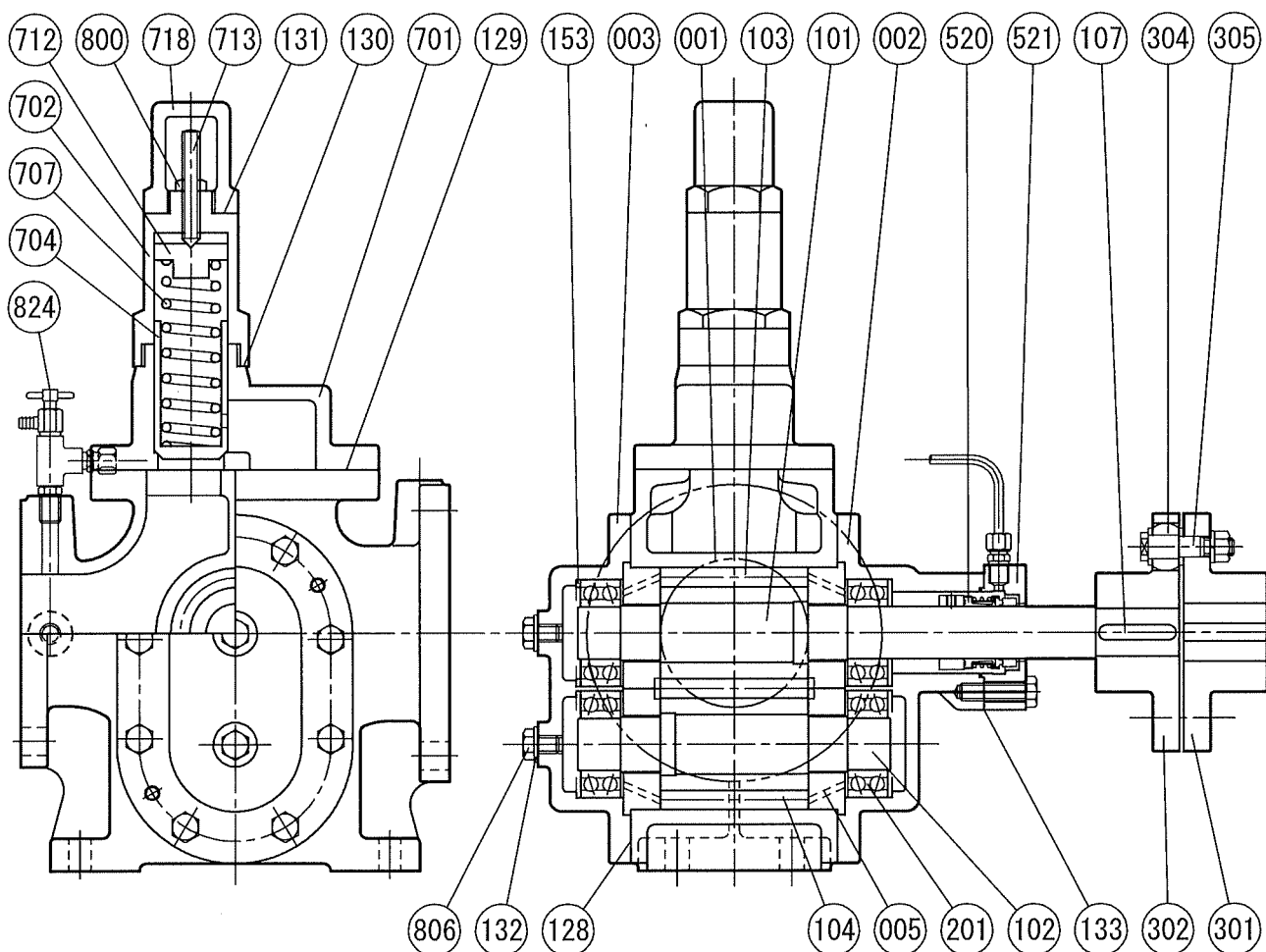
MODEL ALG-65

DRAWING NO.

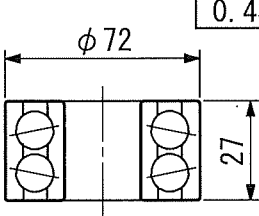
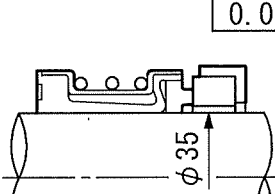
CODS0132

DATE

4 JUN. 2020



201	BALL BEARING		SUJ	4					
153	COLLAR	CARBON STEEL	S45C	1	824	AIR COCK	COPPER ALLOY ROD	C3604	1
133	GASKET	NON ASBESTOS FIBER		1	806	BOLT PLUG	MILD STEEL	SS400	2
132	GASKET	NON ASBESTOS FIBER		2	800	ADJUST NUT	MILD STEEL	SS400	1
131	GASKET	NON ASBESTOS FIBER		1	718	CAP	CAST IRON	FC200	1
130	GASKET	NON ASBESTOS FIBER		1	713	ADJUST SCREW	MILD STEEL	SS400	1
129	GASKET	NON ASBESTOS FIBER		1	712	SPRING CARRIER	MILD STEEL	SS400	1
128	GASKET	NON ASBESTOS FIBER		2	707	VALVE SPRING	PIANO WIRE	SWP	1
107	COUPLING KEY	CARBON STEEL	S45C	1	704	SAFETY VALVE	CARBON STEEL	S45C	1
104	DRIVEN GEAR	CARBON STEEL	S45C	1	702	VALVE COVER	CARBON STEEL	S45C	1
103	DRIVE GEAR	CARBON STEEL	S45C	1	701	SAFETY VALVE BOX	CAST IRON	FC200	1
102	DRIVEN SHAFT	CARBON STEEL	S45C	1	521	SEAL COVER	CAST IRON	FC200	1
101	DRIVE SHAFT	CARBON STEEL	S45C	1	520	MECHANICAL SEAL			1 SET
005	SIDE PLATE	CAST IRON	FC200	4	305	COUPLING BOLT & NUT	MILD STEEL	SS400	1 SET
003	SIDE COVER	CAST IRON	FC200	1	304	COUPLING RING	RUBBER		1 SET
002	SIDE COVER	CAST IRON	FC200	1	302	COUPLING	CAST IRON	FC200	1
001	CASING	CAST IRON	FC200	1	301	COUPLING	CAST IRON	FC200	1
PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY

SPARE PARTS LIST							PAGE		8	
LUBE OIL TRANSFER PUMP HORIZONTAL GEAR ROTARY PUMP MODEL: ALG-65 1 SET							SHIP NO.			
							BOX NO.			
Item NO.	NAME	SKETCH	MASS : kg	MATERIAL	SUPPLY		DRAWING NO.		REMARKS	
					WORK-ING PER PUMP	SPARE PER SHIP	CODS0132			
							NO.	PART NO.		
1	BALL BEARING		0.43	SUJ	4	1		201		
2	MECHANICAL SEAL		0.09		1 SET	1 SET		520		
MFR' S NAME & ADDRESS			 NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN					DRAWING NO.		
								A1DP0715		



SECTIONAL DRAWING

HORIZONTAL GEAR ROTARY PUMP

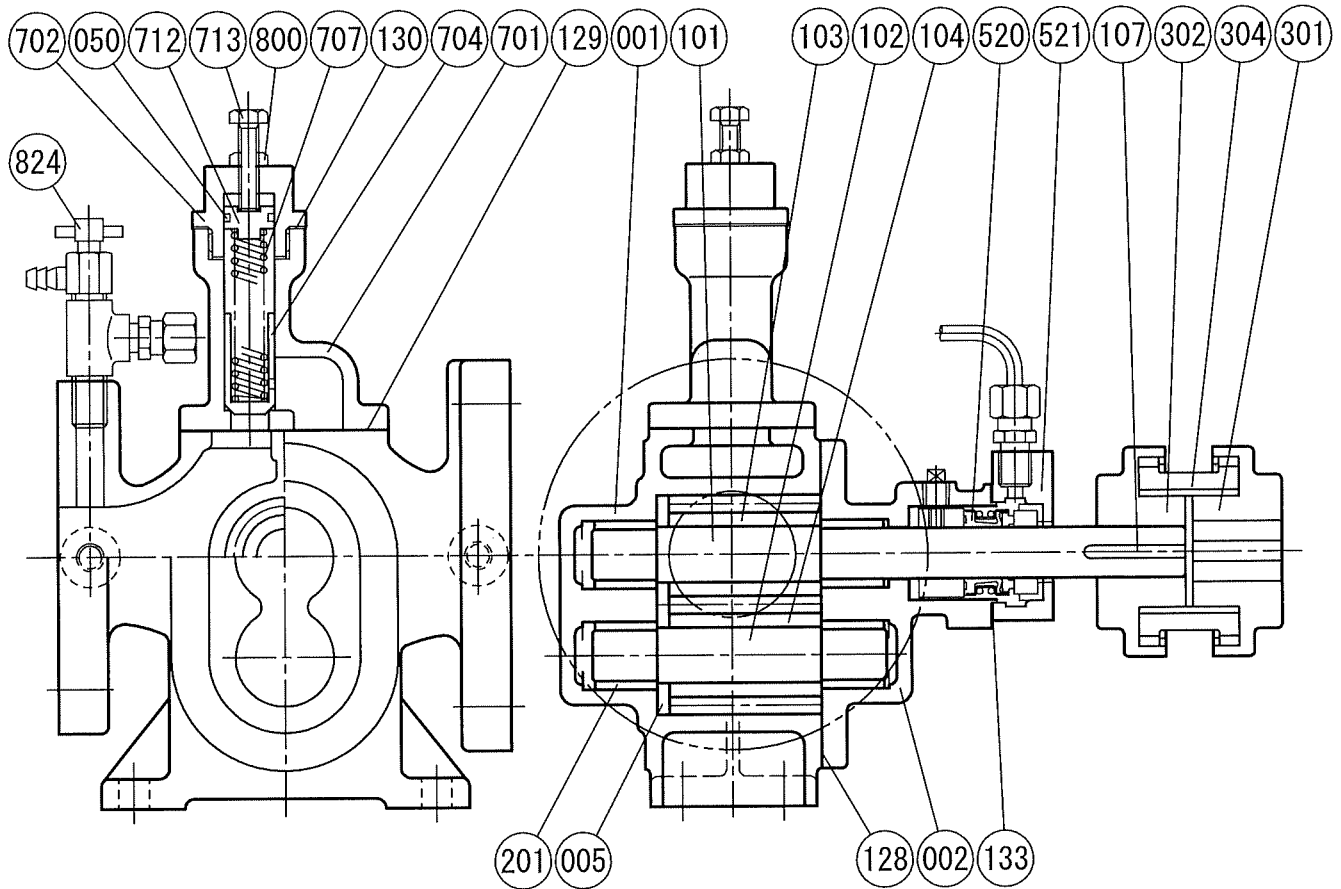
MODEL : ALG-N

DRAWING NO.

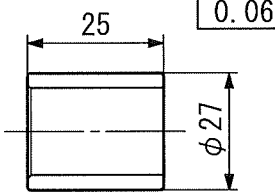
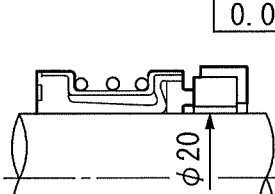
B6DS0215

DATE

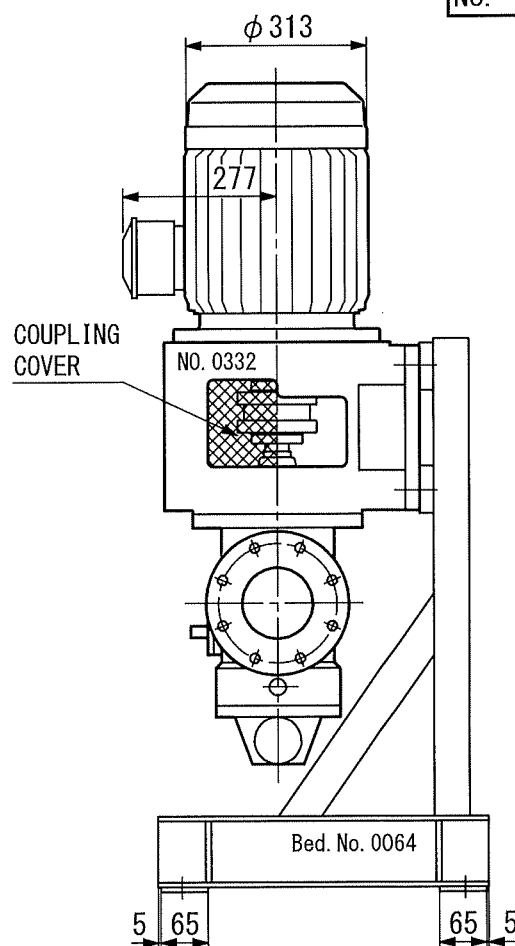
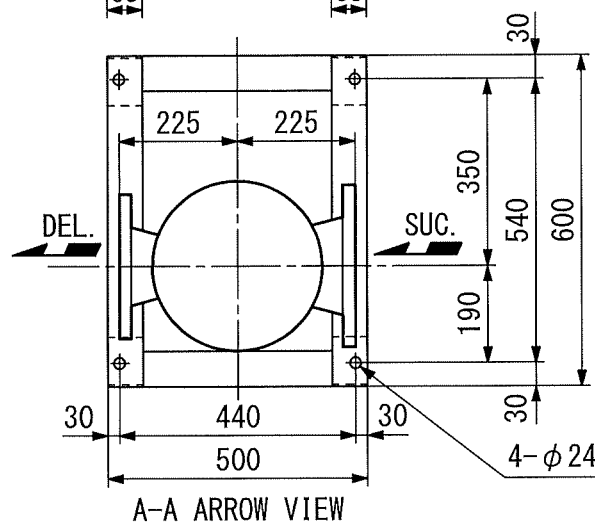
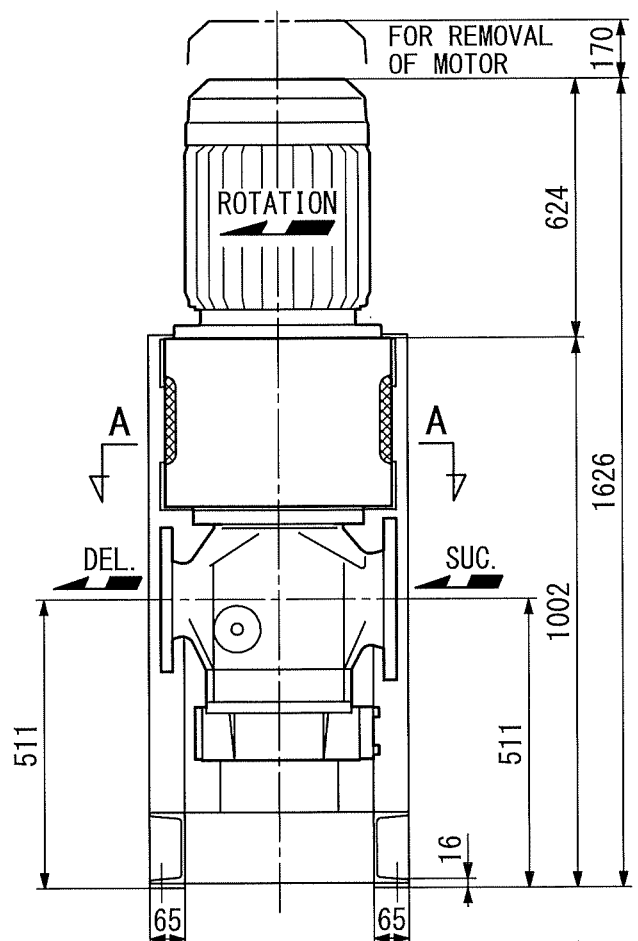
26 AUG. 2016



201	BEARING METAL	LEAD BRONZE	CAC603	4					
133	GASKET	NON ASBESTOS FIBER		1	824	AIR COCK	COPPER ALLOY ROD	C3604	1
130	GASKET	NON ASBESTOS FIBER		1	800	ADJUST NUT	MILD STEEL	SS400	1
129	GASKET	NON ASBESTOS FIBER		1	713	ADJUST SCREW	MILD STEEL	SS400	1
128	GASKET	NON ASBESTOS FIBER		1	712	SPRING CARRIER	MILD STEEL	SS400	1
107	COUPLING KEY	CARBON STEEL	S45C	1	707	VALVE SPRING	PIANO WIRE	SWP	1
104	DRIVEN GEAR	CARBON STEEL	S45C	1	704	SAFETY VALVE	CARBON STEEL	S45C	1
103	DRIVE GEAR	CARBON STEEL	S45C	1	702	VALVE COVER	CAST IRON	FC200	1
102	DRIVEN SHAFT	CARBON STEEL	S45C	1	701	SAFETY VALVE BOX	CAST IRON	FC200	1
101	DRIVE SHAFT	CARBON STEEL	S45C	1	521	SEAL COVER	CAST IRON	FC200	1
050	O-RING	RUBBER		1	520	MECHANICAL SEAL			1 SET
005	SIDE PLATE	CAST IRON	FC200	2	304	COUPLING SLEEVE	RUBBER		1
002	SIDE COVER	CAST IRON	FC200	1	302	COUPLING	CAST IRON	FC200	1
001	CASING	CAST IRON	FC200	1	301	COUPLING	CAST IRON	FC200	1
PART NO.	NAME OF PART	MATERIAL		Q' TY	PART NO.	NAME OF PART	MATERIAL		Q' TY

SPARE PARTS LIST							PAGE		9		
NO. 1 & NO. 2 M/E CYLINDER OIL TRANSFER PUMP							HORIZONTAL GEAR ROTARY PUMP MODEL: ALG-32N 2 SETS				
							SHIP NO.				
							BOX NO.				
Item NO.	NAME	SKETCH	MASS : kg	MATERIAL	SUPPLY		DRAWING NO.		REMARKS		
					WORK- ING PER PUMP	SPARE PER SHIP	NO.	PART NO.			
1	BEARING METAL		0.06	LEAD BRONZE	4	1		201			
2	MECHANICAL SEAL		0.05		1 SET	1 SET		520			

MFR' S NAME & ADDRESS	 NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN	DRAWING NO.
		98DP0191



SUC. & DEL. FLANGE

DIN PN16	DIN PN16
SUC.	DEL.
φ 150	φ 125
φ 240	φ 210
φ 285	φ 254
8-φ 22	8-φ 18

MASS	
PUMP	292 kg
MOTOR	150 kg
TOTAL	442 kg

ACCESSORIES FOR EACH 1 PUMP		SPECIFICATION		PUMP NAME	
DESCRIPTION	Q' TY	SUCTION BORE	150 mm	NO. 1 & NO. 2 M/E T/C L. O. PUMP	
COUPLING & COVER	1set	DELIVERY BORE	125 mm		
SAFETY VALVE (SET PRESSURE 0.33 MPa)	1set	CAPACITY	32 m ³ /h	PUMP MODEL	ALTV-90
PRESSURE GAUGE WITH JOINT	1set	TOTAL PRESSURE	0.28 MPa	QUANTITY	2
COMPOUND GAUGE WITH JOINT	1set	SUCTION PRESSURE	-0.05 MPa	RULE	LR
GAUGE ROOT VALVE	2	REVOLVING SPEED	1150 min ⁻¹	ANGLE PROJ.	3 rd
GAUGE BOARD	1	HYD. TEST PRESS.	0.7 MPa	SCALE	FREE
COUNTER FLANGE	1set	MOTOR OUTPUT	11kW , 6P	DRAWING NO.	C2DG0273
		POWER SOURCE	440V, 60Hz, 3ph	 NANIWA PUMP MFG.CO.,LTD. DESIGN SECTION TECHNICAL DEPARTMENT	
		GUARANT. VIS. FOR POWER	1000 mm ² /s		
		GUARANT. VIS. FOR CAPA.	26 mm ² /s		
		OIL TEMP =	60 °C		



SECTIONAL DRAWING

VERTICAL 3-SCREW PUMP

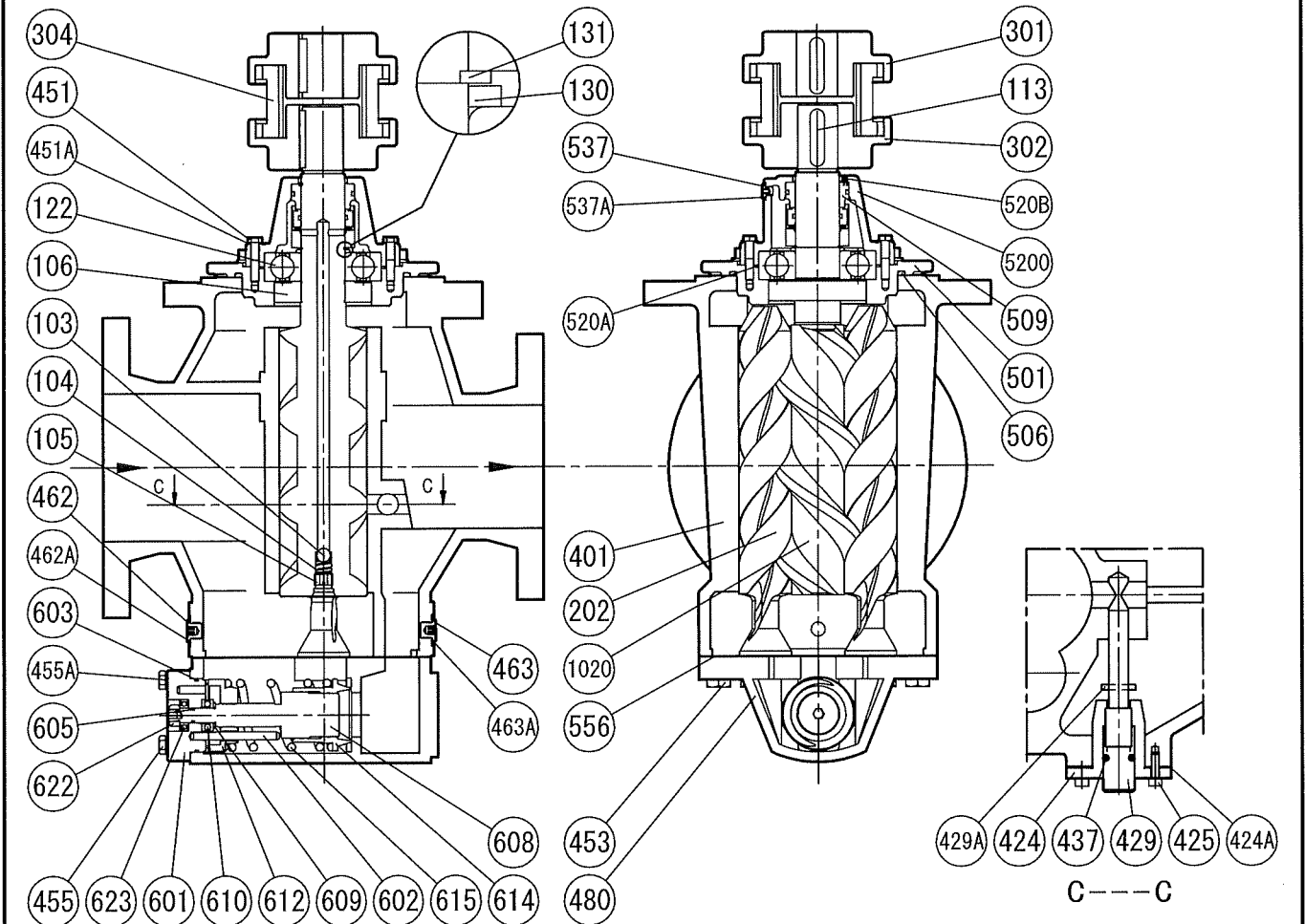
MODEL : ALT V-80/90

DRAWING NO.

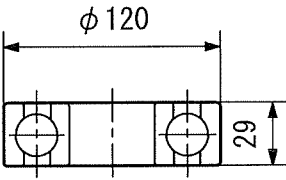
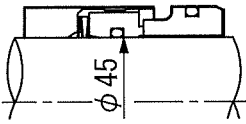
B6DS0100

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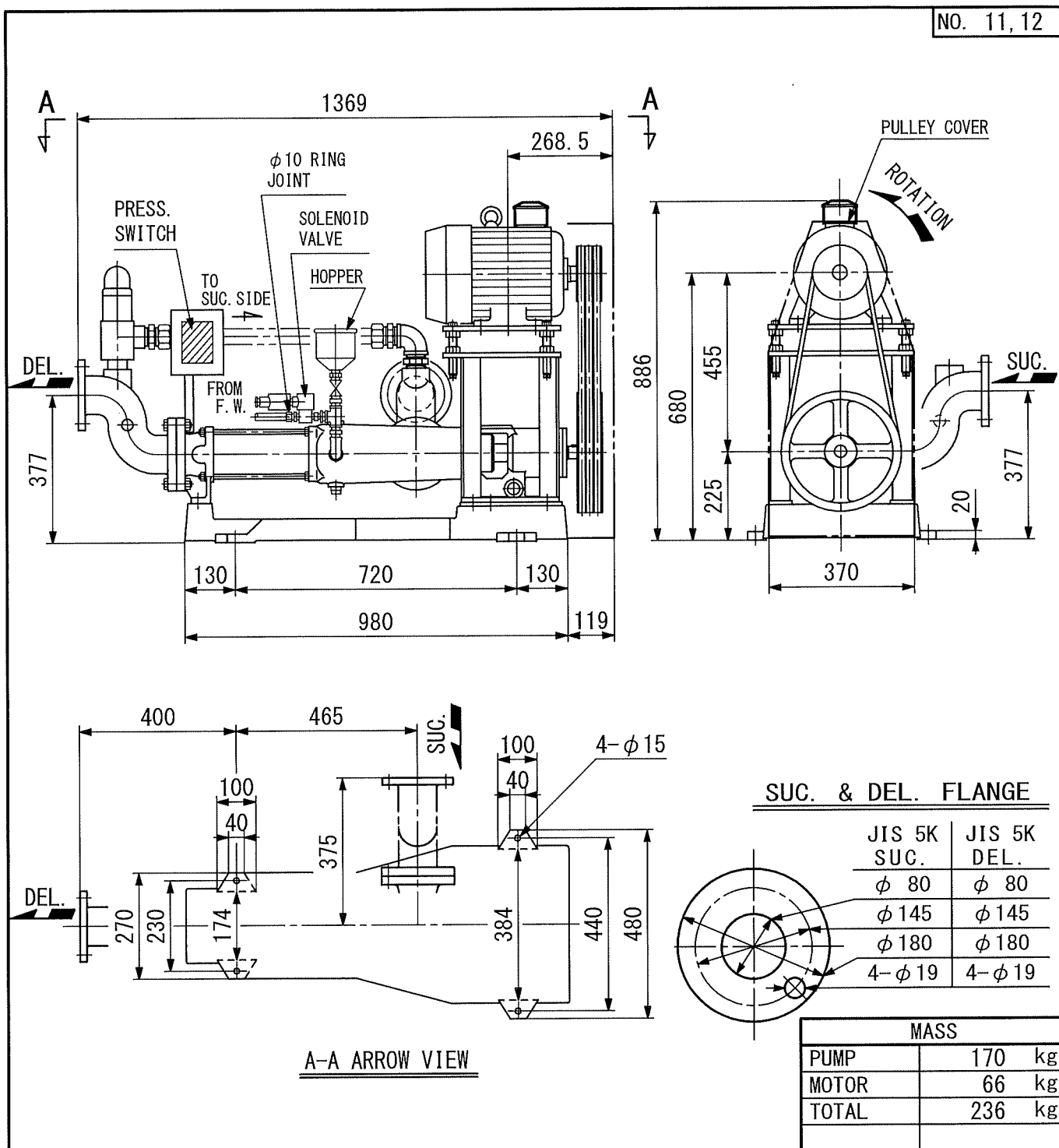
25 MAR. 2016



462	PLUG	MILD STEEL	SS400	1					1
455A	WASHER	MILD STEEL	SS400	4	623	BALL BEARING	SUJ		1
455	SCREW	MILD STEEL	SS400	4	622	NUT	MILD STEEL	SS400	1
453	SCREW	MILD STEEL	SS400	8	615	VALVE SPRING	CARBON STEEL	S45C	1
451A	WASHER	MILD STEEL	SS400	4	614	VALVE PISTON	CARBON STEEL	S45C	1
451	SCREW	MILD STEEL	SS400	4	612	REGULATING NUT	MILD STEEL	SS400	1
437	O RING	RUBBER		1	610	BALL BEARING	SUJ		1
429A	TENSION PIN	CARBON STEEL	S45C	1	609	WASHER	MILD STEEL	SS400	1
429	GUIDING SCREW	CARBON STEEL	S45C	1	608	VALVE SPINDLE	MILD STEEL	SS400	1
425	SCREW	MILD STEEL	SS400	4	605	O RING	RUBBER		1
424A	GASKET	KLINGER SIL		1	603	O RING	RUBBER		1
424	COVER	CARBON STEEL	S45C	1	602	PIN	CARBON STEEL	S45C	1
401	PUMP BODY	CAST IRON	FC200	1	601	VALVE COVER	DUCTILE CAST IRON	FCD400	1
304	COUPLING SLEEVE	RUBBER		1	556	GASKET	KLINGER SIL		1
302	COUPLING	CAST IRON	FC200	1	537A	WASHER	ALUMINIUM		1
301	COUPLING	CAST IRON	FC200	1	537	PLUG	MILD STEEL	SS400	1
202	IDLER ROTOR	STRUCTURAL STEEL	ENT10025-2 S450JO	2	520B	TENSION PIN	STEEL		1
131	RETAINING RING	CARBON STEEL	S45C	1	520A	O RING	RUBBER		1
130	SUPPORT RING	CARBON STEEL	S45C	1	5200	COMPL. COVER	DUCTILE CAST IRON	FCD400	1
122	BALL BEARING	SUJ		1	509	MECHANICAL SEAL			1SET
113	KEY	CARBON STEEL	S45C	1	506	O RING	RUBBER		1
106	BALANCING PISTON	CARBON STEEL	S45C	1	501	FRONT COVER	DUCTILE CAST IRON	FCD400	1
105	HOLE SCREW	MILD STEEL	SS400	1	480	VALVE HOUSING	DUCTILE CAST IRON	FCD400	1
104	SPRING	CARBON STEEL	S45C	1	463A	WASHER	ALUMINIUM		1
103	BALL	CARBON STEEL	S45C	1	463	PLUG	MILD STEEL	SS400	1
1020	COMPL. POWER ROTOR	STRUCTURAL STEEL	ENT10025-2 S450JO	1	462A	WASHER	ALUMINIUM		1
PNO.	NAME OF PART	MATERIAL		Q' TY	PNO.	NAME OF PART	MATERIAL		Q' TY

SPARE PARTS LIST							PAGE		10		
NO. 1 & NO. 2 M/E T/C L. O. PUMP							VERTICAL 3-SCREW PUMP MODEL: ALTV-90 2 SETS				
							SHIP NO.				
							BOX NO.				
Item NO.	NAME	SKETCH	MASS : kg	MATERIAL	SUPPLY		DRAWING NO.		REMARKS		
					WORK- ING PER PUMP	SPARE PER SHIP	B6DS0100 NO.	PART NO.			
1	BALL BEARING		1.6	SUJ	1	1		122			
2	MECHANICAL SEAL		0.4		1 SET	1 SET		509			

MFR' S NAME & ADDRESS	 NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN	DRAWING NO.
		B6DP0025



ACCESSORIES FOR EACH 1 PUMP		SPECIFICATION		PUMP NAME	
DESCRIPTION	Q' TY	SUCTION BORE	80 mm	GREY WATER DISCHARGE PUMP SEWAGE DISCHARGE PUMP	
V-BELT & PULLEY & COVER	1set	DELIVERY BORE	80 mm		
COMMON BED	1	CAPACITY	10 m ³ /h	PUMP MODEL	ANH-50
SAFETY VALVE (SET PRESSURE 0.25 MPa)	1set	TOTAL PRESSURE	0.2 MPa	QUANTITY	TOTAL 2
HOPPER & VALVE	1set	SUCTION PRESSURE	-0.05 MPa	RULE	LR
PRESSURE GAUGE WITH JOINT	1set	REVOLVING SPEED	520 min ⁻¹	ANGLE PROJ.	3 rd
COMPOUND GAUGE WITH JOINT	1set	HYD. TEST PRESS.	0.5 MPa	SCALE	FREE
GAUGE ROOT VALVE & BOARD	2	MOTOR REV. SPEED	1150 min ⁻¹	DRAWING NO.	B4DG0617
SOLENOID VALVE	1set	MOTOR OUTPUT	3.7kW , 6P	 NANIWA PUMP MFG.CO.,LTD. DESIGN SECTION TECHNICAL DEPARTMENT	
PIECE (SUC. /DEL.)	1set	POWER SOURCE	440V, 60Hz, 3ph		
PRESS. SWITCH ON :0.15 MPa OFF:0.05 MPa	1	WORKING TEMP = 50 °C			
COUNTER FLANGE	1set				